One Causal Seed for the Entire Rényi Entropy Spectrum

Research note for the finite-Shannon-theorem project

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Abstract

Assuming the finite refined-cutoff bound of the project, we construct one exact countable causal seed on a locally finite controlled tree whose Rényi entropy is within an explicit additive constant of the ordinary, possibly history-inconsistent optimum, simultaneously at every positive order. No finite Shannon entropy assumption is needed. The same seed attains the exact min-entropy optimum. The main step is a distributional compactness argument: one limiting seed retains the finite stochastic sandwich, and therefore all entropy comparisons at once.

Status. This is a mathematical derivation from the finite results in [1], not a new Lean certificate. The infinite simultaneous statement below has not been independently reviewed. No claim of historical priority or sharpness of its constants is made.

1 Model and statement

Fix a rooted controlled history tree with finite nonempty action and output sets at every history and strictly positive transition probabilities. Finite horizons therefore have finitely many histories and policies. Terminal histories may be padded by deterministic steps.

A *causal seed* is a countable probability law on total response strategies: a strategy specifies an output at every history–action pair. It is exact if it realizes the prescribed law under every deterministic adaptive policy at every finite horizon. An *ordinary source* is a countable random variable with a deterministic readout for each such policy and horizon; agreement between the readouts at shared histories is not required. Assume at least one ordinary exact source exists.

All logarithms and entropies are in bits. Write

$$O_\alpha = \inf_{U \text{ ordinary exact}} H_\alpha(U), \qquad C_\alpha = \inf_{U \text{ causal exact}} H_\alpha(U),$$

where $H_\alpha(p) = (1-\alpha)^{-1} \log_2 \sum_i p_i^\alpha$ for $\alpha \neq 1$, H_1 is Shannon entropy, and $H_\infty(p) = -\log_2 \sup_i p_i$. Infinite values are allowed.

Theorem 1. *There exists a single exact countable causal seed S such that, simultaneously for every $0 < \alpha < \infty$,*

$$\boxed{O_\alpha \le C_\alpha \le H_\alpha(S) \le O_\alpha + d_\alpha,}$$

where

$$d_\alpha = \frac{\log_2\left(\frac{1}{2} + \frac{2^{-\alpha}}{\alpha}\right)}{1-\alpha} \quad (\alpha \neq 1), \qquad d_1 = \frac{1 + \log_2 e}{2}.$$

Moreover $H_\infty(S) = C_\infty = O_\infty$. Here ∞ denotes the Rényi order, not the time horizon.

2 The finite input and its infinite limit

For the horizon-n transcript law $P_{n,\pi}$, put

$$F_n(t) = \min_{\pi} P_{n,\pi}\{-\log_2 P_{n,\pi}(X) \le t\}, \qquad F(t) = \inf_n F_n(t).$$

Zero-probability transcripts carry no mass in this expression. Extending a policy and refining its transcript shows $F_{n+1} \le F_n$. Each F_n is a CDF; F is nondecreasing and right-continuous. For the latter, use $F(t) = \inf_n \inf_{u>t} F_n(u) = \inf_{u>t} F(u)$.

Let $I_U = -\log_2 p_U(U)$ for any countable source. Deterministic readouts merge atoms, so exactness gives

$$F_{I_U}(t) \le F_n(t) \quad \text{for all } n, t. \tag{1}$$

Consequently $F(t) \to 1$ as $t \to \infty$ whenever an ordinary countable source exists. Thus F is the CDF of a finite nonnegative random variable Z. Let Z_n have CDF F_n; then $Z_n \preceq_{\mathrm{st}} Z$ and $Z_n \Rightarrow Z$.

Finite input. The refined greedy cutoff in [1] gives a finite exact causal seed at each horizon, with information J_n, for which

$$Z_n \preceq_{\mathrm{st}} J_n \preceq_{\mathrm{st}} Z_n + W, \qquad W = A(1+E), \quad A \sim \mathrm{Bernoulli}(1/2), \quad E \sim \mathrm{Exp}(\ln 2). \tag{2}$$

Here A, E, Z_n are independent. To be explicit about this input, the repository's `smallWeight_le_envelope` bounds $\mathbb{P}(J_n \ge t)$ by the envelope average of $\psi(2^{Z_n - t})$, where

$$\psi(2^{-u}) = \begin{cases} 1, & u \le 0, \\ 1/2, & 0 < u < 1, \\ 2^{-u}, & u \ge 1. \end{cases}$$

This is $\mathbb{P}(W \ge u)$. Taking thresholds $t + \delta$ with $\delta \downarrow 0$ gives the usual strict-tail inequality in (2), including the atom of W at zero. The lower inequality is the deterministic-readout bound (1).

Lemma 2. *There exists an exact countable causal seed S with information J such that*

$$\boxed{Z \preceq_{\mathrm{st}} J \preceq_{\mathrm{st}} Z + W,}$$

with Z and W independent on the right-hand side.

Proof. List the finite greedy atom weights in decreasing order, pad by zeros, and write them as $p_{n,i}$, $i \ge 1$. Extend each finite response strategy arbitrarily to the full infinite tree. As $Z_n \preceq_{\mathrm{st}} Z$, (2) gives $J_n \preceq_{\mathrm{st}} Z + W$. For $t \ge 0$, at most $\lfloor 2^t \rfloor$ atoms have weight at least 2^{-t}. Hence

$$\sum_{i > \lfloor 2^t \rfloor} p_{n,i} \le \mathbb{P}(J_n > t) \le \mathbb{P}(Z + W > t). \tag{3}$$

This is a uniform tail estimate for the *ranked weights*, with no entropy assumption.

The full strategy space is a countable product of finite discrete sets. Diagonal extraction gives a subsequence $n_j \to \infty$ such that, for every i, $p_{n_j,i} \to p_i$ and the associated strategies converge coordinatewise to s_i. Finite partial sums have limits. By (3), their tails are uniformly arbitrarily small, so $\sum_i p_i = 1$ and $\sum_i |p_{n_j,i} - p_i| \to 0$.

Give label i weight p_i and response strategy s_i, discarding zero weights. Fix a finite horizon, policy and transcript. The indicator that strategy $s_{n_j,i}$ generates that transcript eventually equals its limit for each fixed i, because it depends on finitely many coordinates. Splitting the weighted sum at a finite rank, and then using the uniform tail estimate, proves exactness of the limit. This works for every finite policy law.

For completeness, $J_{n_j} \Rightarrow J$ follows directly from the same argument. For bounded continuous f, write

$$\mathbb{E}f(J_{n_j}) = \sum_i p_{n_j,i} f(-\log_2 p_{n_j,i}).$$

A fixed positive limiting weight gives convergence of its term; a zero limiting weight gives a term tending to zero in absolute value. Uniformly small rank tails control the remaining sum. Thus the expectations converge to $\sum_{i:p_i>0} p_i f(-\log_2 p_i)$.

Finally $Z_{n_j} \Rightarrow Z$ and $Z_{n_j} + W \Rightarrow Z + W$. First-order stochastic order is closed under weak convergence: test against bounded continuous increasing functions. Passing to the limit in (2) proves the lemma. □

3 Simultaneous entropy bounds

Proof of Theorem 1. Take the single seed of Lemma 2. For $\alpha \neq 1$, define

$$R_\alpha(Z) = \frac{1}{1-\alpha} \log_2 \mathbb{E}2^{(1-\alpha)Z}.$$

The stochastic sandwich and independence give

$$R_\alpha(Z) \leq H_\alpha(S) \leq R_\alpha(Z) + \frac{\log_2 \mathbb{E}2^{(1-\alpha)W}}{1-\alpha}. \tag{4}$$

For $0 < \alpha < 1$ these follow by monotonicity of the increasing exponential; infinite moments are permitted. For $\alpha > 1$ the exponential is decreasing, and division by $1-\alpha < 0$ reverses the inequality once more. Its expectation is strictly positive because Z is finite almost surely.

Since $1-\alpha < 1$, the exponential moment of W is finite, and

$$\mathbb{E}2^{(1-\alpha)W} = \frac{1}{2} + \frac{1}{2} 2^{1-\alpha} \int_0^\infty (\ln 2)\, 2^{-\alpha x}\, dx = \frac{1}{2} + \frac{2^{-\alpha}}{\alpha}.$$

For $\alpha = 1$, use the nonnegative means directly:

$$\mathbb{E}Z \leq H_1(S) \leq \mathbb{E}Z + \mathbb{E}W, \qquad \mathbb{E}W = \frac{1}{2}\left(1 + \frac{1}{\ln 2}\right) = d_1.$$

This also agrees with the continuous extension of d_α at 1.

For any ordinary exact source U, (1) says $Z \preceq_{\mathrm{st}} I_U$. Therefore $R_\alpha(Z) \leq H_\alpha(U)$, and $\mathbb{E}Z \leq H_1(U)$. Taking the infimum over such U in (4) yields $H_\alpha(S) \leq O_\alpha + d_\alpha$. The other inequalities follow from inclusion of feasible classes and feasibility of S.

For the endpoint, fix any ordinary exact U. The preceding bound gives $H_\alpha(S) \leq H_\alpha(U) + d_\alpha$ for every $\alpha > 1$. For a countable probability law, $H_\alpha \to H_\infty$ as $\alpha \to \infty$, whereas $d_\alpha \to 0$. Consequently $H_\infty(S) \leq H_\infty(U)$ for every U. Taking the infimum and using $O_\infty \leq C_\infty \leq H_\infty(S)$ proves equality. No interchange of an infimum with a limit is needed. □

Corollary 3 (Simultaneous finiteness)**.** *For the same seed S, and for each $0 < \alpha \leq 1$,*

$$H_\alpha(S) < \infty \iff \textit{some ordinary exact countable source has finite } H_\alpha.$$

In particular the set of orders with finite Rényi entropy is realized by one causal source, even when its Shannon entropy is infinite.

Proof. The forward implication uses S itself as an ordinary source. The reverse implication follows from the theorem and the finite constant d_α. □

Examples of the bounds. The same S satisfies

$$H_1(S) \le O_1 + 1.2213475204\ldots, \qquad H_2(S) \le O_2 + \log_2(8/5), \qquad H_\infty(S) = O_\infty.$$

These are simultaneous approximation statements, not exact simultaneous minimization at finite orders. The construction does not claim that S also attains the Shannon optimum.

4 Dependencies and limits of the claim

The finite stochastic input is read from `CausalSeed/Entropy.lean` (in particular `smallWeight_le_envelope` and the deterministic-readout lower bound). The noise profile and constants are discussed in `CausalSeed/Note.lean`. Both belong to the repository snapshot `10c0865`. The compactness and passage of the full stochastic sandwich to infinity are proved above; they are not asserted here to have been checked by Lean.

The assumptions retain strictly positive kernels. This note does not establish sharpness of d_α, compute the universal Shannon causal gap, or claim a finite algorithm for constructing the limiting countable seed. The finite constants, information-spectrum method, and ordinary coupling approximations have antecedents in [2, 3, 4]. A dedicated comparison is required before claiming novelty for the simultaneous history-consistent infinite-horizon formulation.

References

[1] J. Mikulík, *finite-Shannon-theorem*, source repository, snapshot `10c0865`, 11 October 2026. https://github.com/jendamikulik/finite-Shannon-theorem.

[2] *Minimum-Entropy Coupling Approximation Guarantees Beyond the Majorization Barrier*, AISTATS 2023. https://arxiv.org/abs/2302.11838.

[3] *Information Spectrum Converse for Minimum Entropy Couplings and Functional Representations*, 2023. https://arxiv.org/abs/2305.05745.

[4] *Efficient Approximate Minimum Entropy Coupling of Multiple Probability Distributions*, preprint 2020; IEEE Transactions on Information Theory, 2021. https://arxiv.org/abs/2006.07955.